Normal left, right and between probabilities

1 RECOGNISE IT

- Wording says below/less than, above/more than, or between.
- X is normal and μ, σ are supplied.
- The required answer is an area/probability, not a percentile value.

2 DECISION / METHOD

1. Draw the tail(s).
2. Convert boundaries to z or use NORM.DIST directly.
3. Left: CDF. Right: 1-CDF. Between: upper CDF minus lower CDF.
4. Check area direction visually.

3 FORMULA

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Left: P(X<x)=F(x)
Right: P(X>x)=1-F(x)
Between: P(a<X<b)=F(b)-F(a)
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: no normal tail formulas on the one-page sheet.
Still know: CDF direction, complement, and upper-minus-lower ordering.

5 EXCEL ROUTE

- `=NORM.DIST(x,mu,sigma,TRUE)`
- right: `=1-NORM.DIST(x,mu,sigma,TRUE)`
- between: two NORM.DIST calls, upper minus lower.

6 TINY WORKED EXAMPLE

For $X \sim N(50,10^2)$, `P(40<X<60)=NORM.DIST(60,50,10,TRUE)-NORM.DIST(40,50,10,TRUE)=0.6827`.

7 TOP EXAM TRAPS

- Right tail without the 1-minus.
- Between probabilities added instead of subtracted.
- Confusing X with $\bar{X}$; $\bar{X}$ would use SE.
- Course notation uses variance σ^2, while NORM.DIST uses SD σ.

8 SAY ALOUD

Below is the CDF, above is one minus the CDF, and between is upper CDF minus lower CDF.

9 EXIT CHECK

What Excel expression gives $P(X>75)$ for $X \sim N(60,8^2)$?

ONE-GLANCE VISUAL ROUTE

